"More than 50 per cent of the cortex, the surface of the brain, is devoted to processing visual information," said William G. Allyn, a professor of medical optics.

The human visual cortex is approximately made up of around 140 million neurons. In addition to other tasks, this is the key organ responsible for detecting, segmenting and integrating the visual data received by the retina. It is then sent to other regions in the brain where it is processed and analysed to perform object recognition and detection, and subsequently retain it as applicable.

In the early 1950s, David H. Hubel and Torsten Wiesel conducted a series of experiments on cats and won two Nobel prizes for their amazing findings on the structure of the visual cortex. They revolutionised our understanding of the human visual cortex and paved the way for much further research based on this topic.

Their studies showed that in the visual cortex, many neurons have small local receptive fields. A local receptive field means that these neurons react to visual stimuli only if they are located in a specific region of the visual field of perception. In other words, these neurons are fired or activated only if the visual stimulus is located at a particular place on the retina or visual field. They found that some neurons have larger receptive fields and they are fired, activated or react to complex features in the visual field, which in a way are a combination of low-level features to which other neurons react. The low-level features mean horizontal and vertical lines, edges and corners and different angles of lines, while high-level features are the simple or complex combinations of these low-level features. Figure 1 illustrates the local receptive fields of different cells in the human visual cortex.

Hubel and Wiesel also discovered that there are three types of cells in the visual cortex and each has distinct characteristics, based on the features they learn or react to. These are: simple, complex and hyper-complex. Simple cells are responsible for learning basic features like lines and colours; complex cells for learning features like edges, corners and contours, while hyper-complex cells are responsible for learning combinations of features learnt by simple and complex cells. This powerful insight paved the way to understanding how perception is built – the receptive fields of multiple neurons may overlap and together they tile the whole field. These neurons work together, with a few neurons learning features on their own and others learning by combining the features learnt by other neurons, and finally integrating to detect all forms of complex features in any region of the visual field.

For the video of the Hubel and Wiesel experiment, visit *https://www.youtube.com/watch?v=y_l4kQ5wjiw.*

CNN architecture

CNNs are the most preferred deep learning models for image classification or image related problems. Of late, CNNs have also been used to handle problems in fields as diverse as natural language processing, voice recognition, action recognition and document analysis. The most important characteristic of CNNs is that they automatically learn the important features without any guided supervision. Given the images of two classes, for example, dogs and cats, a CNN will be able to learn the distinctive features of dogs and cats by itself.

Compared to other deep learning models for image related problems, CNNs are computationally efficient, which makes them the preferred option since they can be configured to run on many devices. Ever since the arrival of AlexNet in 2012, researchers have successfully built new CNN architectures that have very good accuracy, with powerful and efficient models. Popular CNN model architectures include VGG-16, Inception models, ResNet-50, Xception and MobileNet.

In general, all CNN models have architectures that are built using the following building blocks. Figure 2 depicts the basic architecture of a CNN model.

- Input layer
- Convolution layers
- Activation functions
- Pooling layers
- Dropout layers
- Fully connected layers
- Softmax layer

In the CNN architecture, the building blocks involving the convolution layer, activation function, pooling layer and dropout layer are repeated before ending with one or more fully connected layer and the softmax layer.

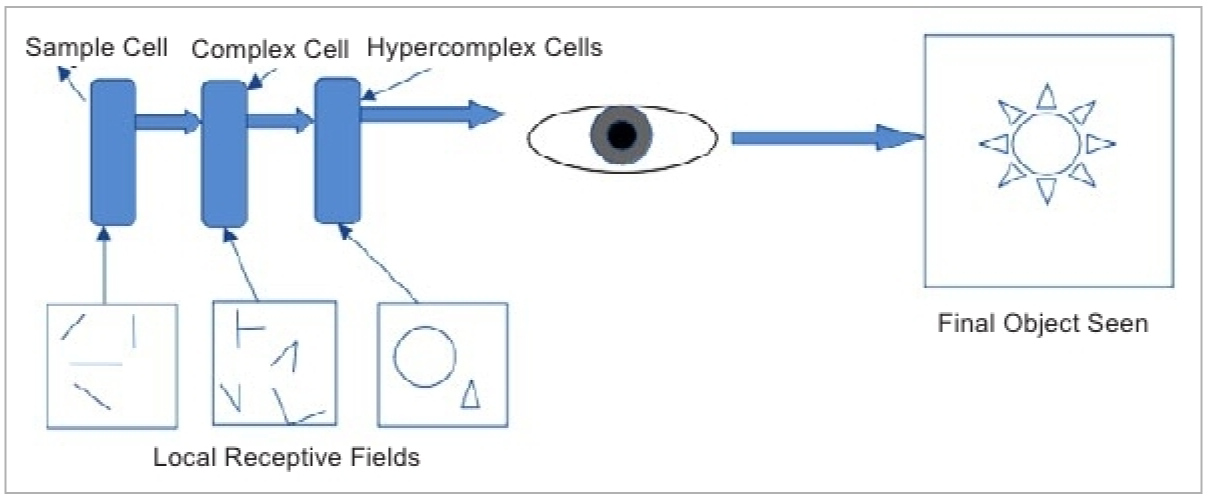

Figure 1: Illustration of local receptive fields and cells in the human visual cortex